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B.Tech. Degree II Semester Regular Examination July 2024

EE/EC/CS/IT 23-200-0201B LINEAR ALGEBRA AND TRANSFORM TECHNIQUES
(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Solve linear system of equations and to determine Eigen values and vectors of a matrix.

CO2: Exemplify the concept of vector space and subspace.

CO3: Determine Fourier series expansion of functions and transform.

CO4: Solve linear differential equation and integral equation using Laplace transform.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

(5 × 2 = 10)

	Marks	BL	CO	PO
I. (a) Find the characteristic equation of $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.	2	L1	1	2
(b) Show that $\{(1,0), (0,1)\}$ forms a basis for the vector space R^2 over R .	2	L1	2	2
(c) Is the set of all real numbers R over the set of all complex numbers C , a vector space? Justify your answer.	2	L2	2	2
(d) Show that $F_s \{x f(x)\} = -\frac{d}{ds} F_c \{f(x)\}$.	2	L1	3	2
(e) Find $L\{t \sin 5t\}$.	2	L2	4	2

PART B

(4 × 10 = 40)

II. (a) Solve the system of equations $2x+3y+4z=31, x-2y+4z=4, 3x+y-3z=2, 6x+2y-6z=4$.	5	L3	1	2
(b) Find the eigen values and eigen vectors of $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$.	5	L4	1	2

OR

III. (a) Find l and m if the rank of the matrix $A = \begin{pmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & l & m \\ 1 & 1 & -2 & l \end{pmatrix}$ is 2.	5	L5	1	2
(b) Using Cayley Hamilton theorem find the inverse of the matrix. $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$	5	L4	1	2

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) Determine whether the vectors (1, 3, 2), (1, -7, 8) and (2, 11, -1) are linearly independent.	5	L3	2	2
	(b) Let V be the vector space of set of all 2×2 matrices. Prove that $\dim(V) = 4$.	5	L4	2	2
OR					
V.	(a) Find an orthonormal basis for the subspace W of the vector space R^3 over the field R generated by the vectors (2, 1, 1) and (1, 3, -1).	5	L5	2	2
	(b) Let V and W be two vector spaces over the same field F and let T be a linear transformation from V to W, prove that Ker T is a subspace of V and Im T is a subspace of W.	5	L4	2	2
VI.	(a) Find the Fourier series expansion for $f(x) = \begin{cases} x + x^2, & -\pi < x < \pi \\ \pi^2, & x = \pm\pi \end{cases}$.	5	L3	3	2
	Hence prove that $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$				
	(b) Find the Fourier integral representation of $f(x) = \begin{cases} 1, & x < 1 \\ 0, & x > 1 \end{cases}$.	5	L6	3	2
	Hence prove that $\int_0^\infty \frac{\cos x \sin x}{x} dx = \frac{\pi}{4}$.				
OR					
VII.	(a) Find the Fourier transform of $e^{-a^2 x^2}$, $a > 0$, hence deduce that $e^{-\frac{x^2}{2}}$ is self reciprocal with respect to Fourier transform.	5	L5	3	2
	(b) Find the Fourier sine transform of $e^{- x }$. Hence evaluate $\int_0^\infty \frac{x \sin mx}{1+x^2} dx$.	5	L6	3	2
VIII.	(a) Evaluate $\int_0^\infty \frac{e^{-4t} \sin^2 t}{t} dt$ using Laplace transform.	5	L5	4	2
	(b) Find the inverse Laplace transform of $\frac{5s^2 - 15s - 11}{(s+1)(s-2)^3}$.	5	L3	4	2
OR					
IX.	(a) State Convolution theorem and evaluate $L^{-1} \left\{ \frac{1}{(s^2+1)(s^2+9)} \right\}$.	5	L4	4	2
	(b) Solve $y'' - 3y' + 2y = e^{2x}$, $y(0) = -3$, $y'(0) = 5$.	5	L6	4	2

Bloom's Taxonomy Levels

L1 - 14.3%, L2 - 9.5%, L3 - 19%, L4 - 23.9%, L5 - 19%, L6 - 14.3%.

